

Manuscript_1

Code and analysis pipeline for: *[Passive Physiological Responses Fail to Predict Context-Dependent Action Selection in Bats]*

Data availability: Raw electrophysiology recordings and behavioral tracking files are deposited at [Zenodo/Dryad — insert DOI]. See [Data](#) below for details.

Overview

This repository contains all code used to analyze bat electrophysiology and behavioral data, including:

- An in vivo electrophysiology pipeline (spike sorting → response analysis → selectivity classification)
- Heart rate analysis during vocalization playback
- Video-based behavioral tracking analysis (approaches, time near speaker, latency)
- Statistical tests and publication-ready figure generation

Subjects were adult bats (*R. aegyptiacus*) tested across four vocalization conditions: **Aggression**, **Distress**, **Echolocation**, and **Distortion** for passive listening recordings. For testing active listening cross contexts, **Distress** and **White Noise** were utilized in playback experiments. Animals were split by sex (male/female) and tested on Day 1 and Day 2 sessions.

Repository Structure

```
Manuscript_1/
├─ Ephys.Pipeline.Final.ipynb      # Main ephys pipeline (Scripts 1–5 + supplementary cells)
├─ BF_Heatmap_Final.ipynb          # Best frequency heatmap (binary BF visualization)
├─ NonparametricNEWKEEP.R          # Heart rate statistical analysis (R)
├─ Hr.Example.py                   # Heart rate timeseries plots (per stimulus, by sex)
├─ RealRadar.Nonpara.py            # Heart rate radar chart + nonparametric stats
├─ BORISGraphs.py                  # Behavioral bout analysis from BORIS annotations
├─ NEW_ST.py                        # Speaker approaches + time near speaker (4 conditions)
├─ ST_overview.py                  # Two-part approach analysis (proportion + metrics)
├─ SpeakerTestHeatMap.py           # KDE position heatmaps across conditions
└─ approaches_day1d2.py            # Day 1 vs Day 2 approach comparison
```

```
└─ xpos_ST.py # Directional movement analysis (toward/away speaker)
```

Data

Data types

Type	Format	Description
Electrophysiology recordings	.npz (continuous)	Raw spike-sorted continuous traces, one file per animal
Ephys per-call output	.npz	Produced by Script 1; contains spike rates, waveforms, timestamps per call
Ephys category summaries	.npz	Produced by Script 2; averaged per animal × vocalization category
Heart rate timeseries	.csv	One file per bat per session; columns: Time, Heart_Rate
Behavioral tracking	.csv	DeepLabCut/tracking output; columns include frame-by-frame x, y position
BORIS behavioral annotations	.csv	Ethogram export with Time, Behavior, Behavior type (START/STOP)

Data access

Raw data are deposited at **[Zenodo]**: [\[DOI link\]](#)

Expected local directory structure (update paths in each script before running):

```
data/
├─ ephys/
│   ├─ continuous/ # Raw .npz continuous recording files (one per animal)
│   └─ processed/ # Auto-generated by Ephys.Pipeline.Final.ipynb
│       ├─ per_call/ # Script 1 output
│       └─ category_avg/ # Script 2 output
├─ heart_rate/
│   ├─ Females/ # HR .csv files for female bats
│   └─ Males/ # HR .csv files for male bats
├─ tracking/ # DeepLabCut tracking .csv files
└─ boris/ # BORIS annotation .csv files
```

Usage

1. Electrophysiology pipeline — `Ephys.Pipeline.Final.ipynb`

Run cells in order. Each cell is a self-contained script:

Script	What it does	Input	Output
Script 1	Per-call processor	Continuous .npz files	Per-call .npz files
Script 2	Category averager	Per-call .npz files	Category-average .npz files
Script 3	Heatmap + Random Forest selectivity	Category .npz files	Publication figures
Script 4	Latency + spike jitter analysis	Per-call .npz files	Latency/jitter figures
Script 5	Raster + PSTH + waveform plots	Continuous + per-call .npz	Per-animal figure PDFs
Supp. cells	BF heatmap, sparseness scatter, GLM decoder, violin plots	Category .npz files	Additional figures

Before running: update `INPUT_FILE_PATHS` and output directory paths at the top of each script cell.

2. Best frequency heatmap — `BF_Heatmap_Final.ipynb`

Standalone notebook. Update `INPUT_FILE_PATHS` to point to your ephys data directory. Produces a binary heatmap showing best-frequency tuning across the recorded population.

3. Heart rate analysis

R script (`NonparametricNEWKEEP.R`): primary statistical analysis. Computes robust baseline HR, calculates change scores per stimulus window, runs Kruskal-Wallis + Dunn post-hoc tests, and produces ggplot2 figures. Update `female_folder_path` and `male_folder_path` at the top.

Python scripts: - `Hr.Example.py` — plots mean $\pm$ SEM heart rate timeseries per stimulus type, averaged by sex - `RealRadar.Nonpara.py` — radar chart of HR change per vocalization category + Mann-Whitney U comparisons

4. Behavioral analysis

All scripts share a similar `CONFIG` block at the top. Update `BASE_DIR` to your tracking CSV directory before running.

Script	Analysis	Output
<code>BORISGraphs.py</code>	Behavior bout durations from BORIS annotations	Bar/violin plots per behavior
<code>NEW_ST.py</code>	Approaches + time near speaker across 4 conditions	Bar plots with significance stars

Script	Analysis	Output
ST_overview.py	Proportion of approachers + latency/duration among approachers	Two-panel figure
SpeakerTestHeatMap.py	KDE position heatmaps per condition	2×2 heatmap grid
approaches_day1d2.py	Day 1 vs Day 2 approach counts per condition	Grouped bar plot
xpos_ST.py	Signed x-displacement during stimulus (toward/away from speaker)	Boxplot per condition

Requirements

Python

Tested with Python 3.9+. Install dependencies with:

```
pip install numpy pandas matplotlib seaborn scipy scikit-posthocs statsmodels
```

Full dependency list:

```
numpy
pandas
matplotlib
seaborn
scipy
scikit-posthocs
statsmodels
pathlib      # stdlib
glob         # stdlib
itertools    # stdlib
```

Note: Ephys.Pipeline.Final.ipynb and BF_Heatmap_Final.ipynb require Jupyter.

R

Tested with R v2024.04.0, Install packages with:

```
install.packages(c("ggplot2", "dplyr", "viridis", "ggpubr", "FSA", "lme4", "lmerTest"))
```

Animals

Individual bats are identified by short alphanumeric IDs (e.g., AF4, B07, AEO). Sex assignments are hardcoded in `RealRadar.Nonpara.py` and `Hr.Example.py` — see `MALE_BATS` sets therein. Stimulus timing windows differ for a subset of animals (AF4, AF6, B32, AD6) due to session timing differences; alternative windows are applied automatically in `NonparametricNEWKEEP.R`.

Citation

If you use this code, please cite:

[Author(s)], [Year]. [Title]. *PNAS*. DOI: [insert]

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